
Pregunta

a) (1,25 punts) Calcula les integrals indefinides $\int (x^3 - 2x + 5) dx$ i $\int \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$.

Solució. $\int (x^3 - 2x + 5) dx = \frac{x^4}{4} - x^2 + 5x + k$ i $\int \left(\frac{1}{x} - \frac{1}{x^2}\right) dx = \ln|x| + \frac{1}{x} + k$.

b) (1,25 punts) Calcula $\int_1^e \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$ i $\int_{-1}^1 (x^3 - 2x + 5) dx$.

Solució. $\int_1^e \left(\frac{1}{x} - \frac{1}{x^2}\right) dx = \left[\ln x + \frac{1}{x}\right]_1^e = \left(1 + \frac{1}{e}\right) - (0 + 1) = \frac{1}{e}$.

$\int_{-1}^1 (x^3 - 2x + 5) dx = \left[\frac{x^4}{4} - x^2 + 5x\right]_{-1}^1 = \left(\frac{1}{4} - 1 + 5\right) - \left(\frac{1}{4} - 1 - 5\right) = 10$.